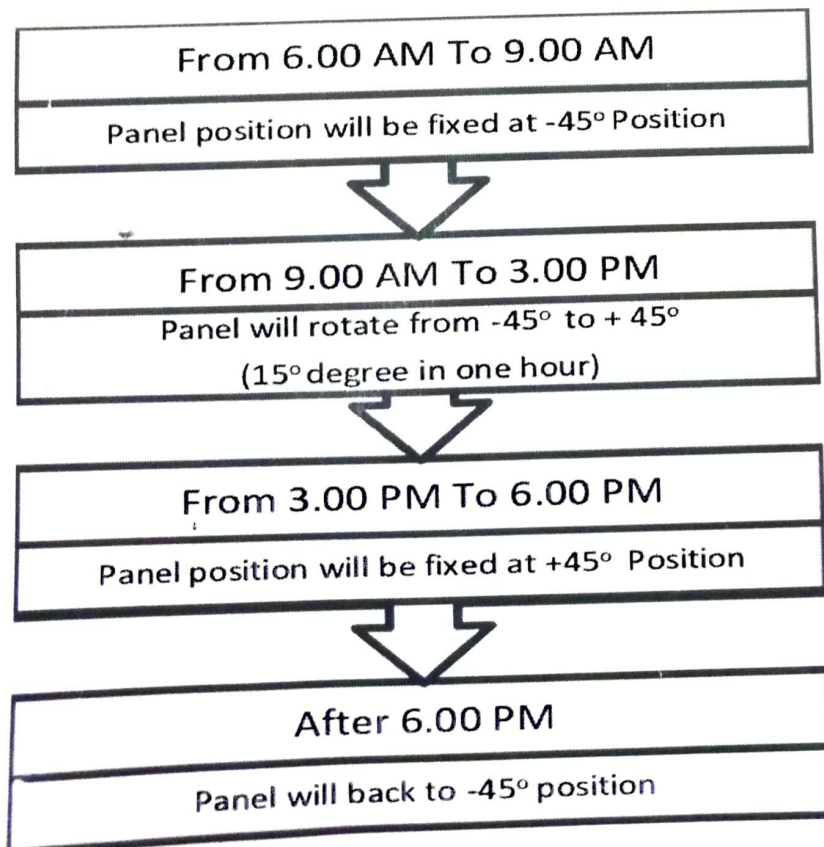
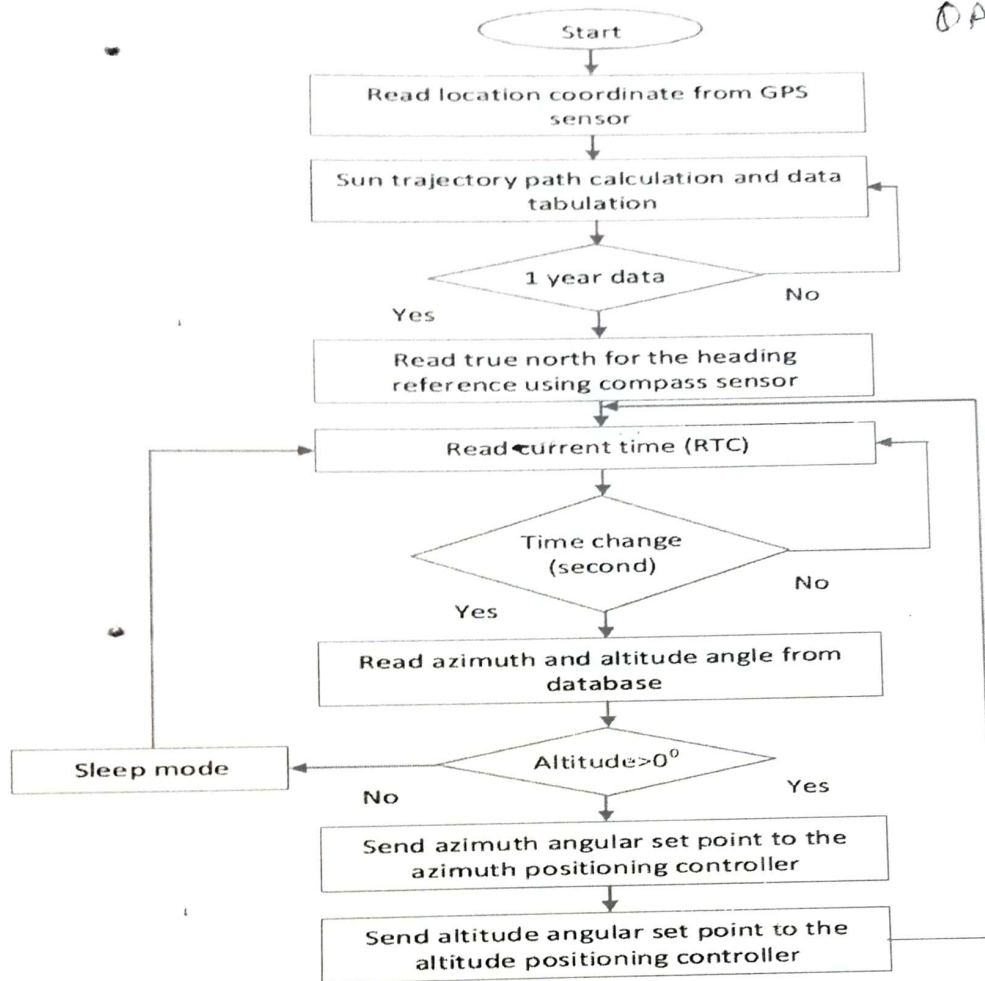


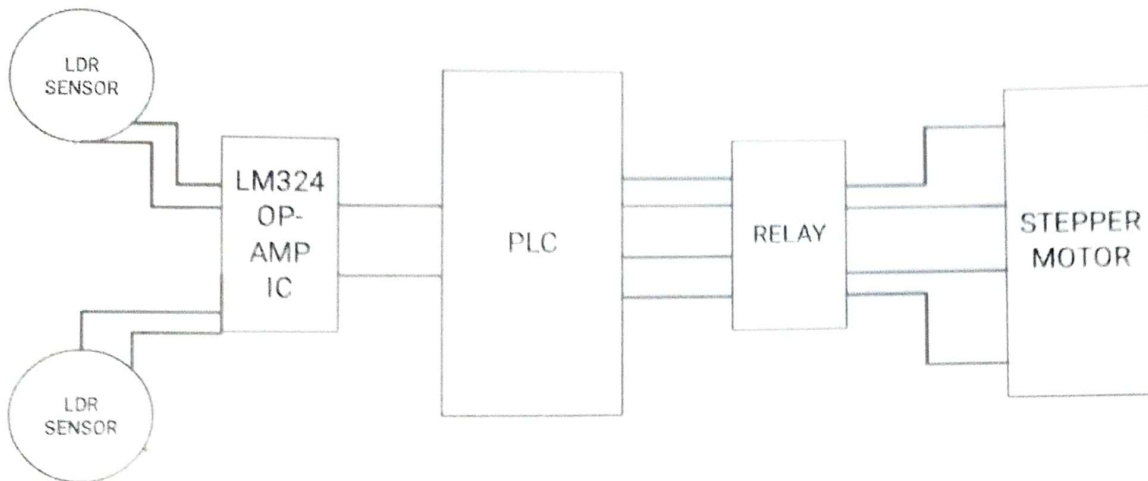
ALGORITHM FOR SOLAR TRACKING

Sun tracking algorithm: This algorithm calculates the solar azimuth and zenith angles of the sun. These angles are then used to position the solar panel or reflector to point toward the sun. Some algorithms are purely mathematical based on astronomical references while others utilize real-time light-intensity readings.

① Active solar tracking



BLOCK DIAGRAM



WORKING PRINCIPLE

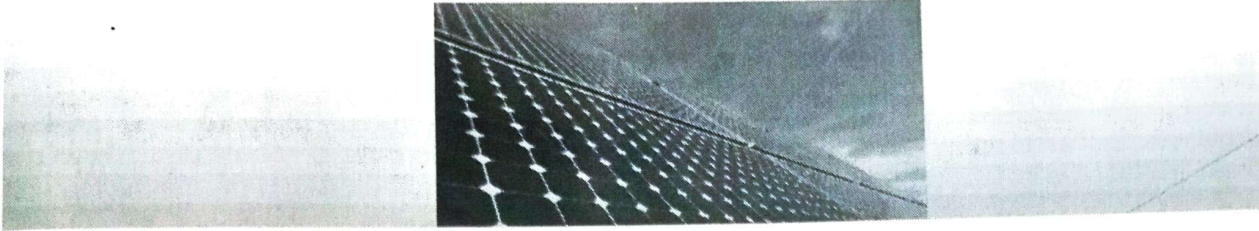
- ❑ Auto-tracking control system composed of PLC, sensors, signal processing units, PV cells, electromagnetic and mechanical motion control modules and power supply systems.
- ❑ Panel detects the sun light strength to sensors.
- ❑ The sensors output is given to the PLC which compares it and produces an equivalent output so as to rotate the motor and align the panel in the direction of sun.

HARDWARE COMPONENTS

1. Solar panel
2. LDR sensors
3. Stepper motor
4. Relay
5. PLC

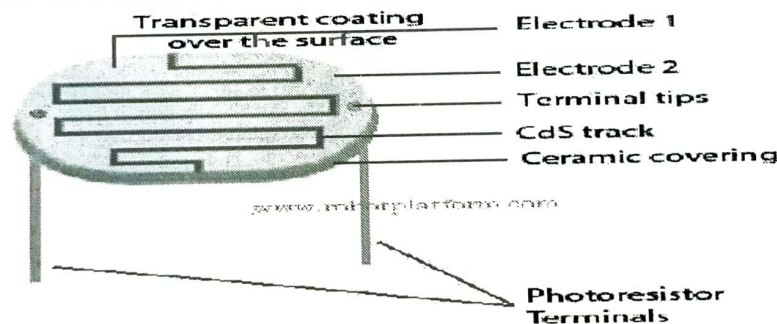
SOLAR PANEL

- Its made up of a number of solar cells connected in series internally
- Convert solar energy to electricity.
- Based on photo voltaic effect.
- Photovoltaic effect produces a change in electrical characteristics when light is incident on it.



LDR SENSORS

- A high resistance semiconductor whose resistance decreases with increasing incident light intensity.
- When light falls on such a semiconductor, the valence electrons receive the light energy incident and jump to the conduction band.
- The conductivity of the device increases, decreasing its resistivity.



STEPPER MOTOR

- Its an electromechanical device which converts digital pulse inputs to incremental shaft rotation.
- Used for precision positioning control applications.
- Its controlled by a series of electromagnetic coils.

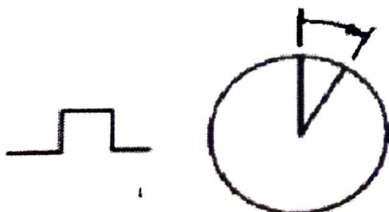


Figure 1: One Pulse Equals One Step

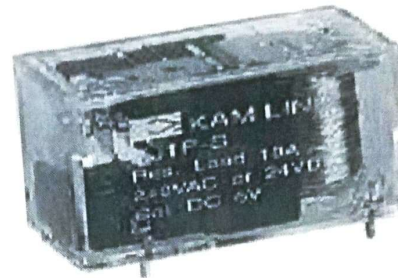
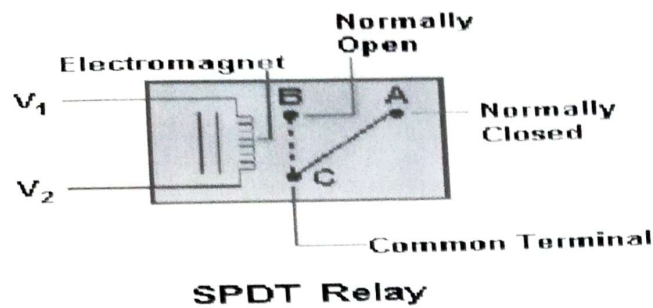


Figure 2: Pulse Count Equals Step Count



RELAY

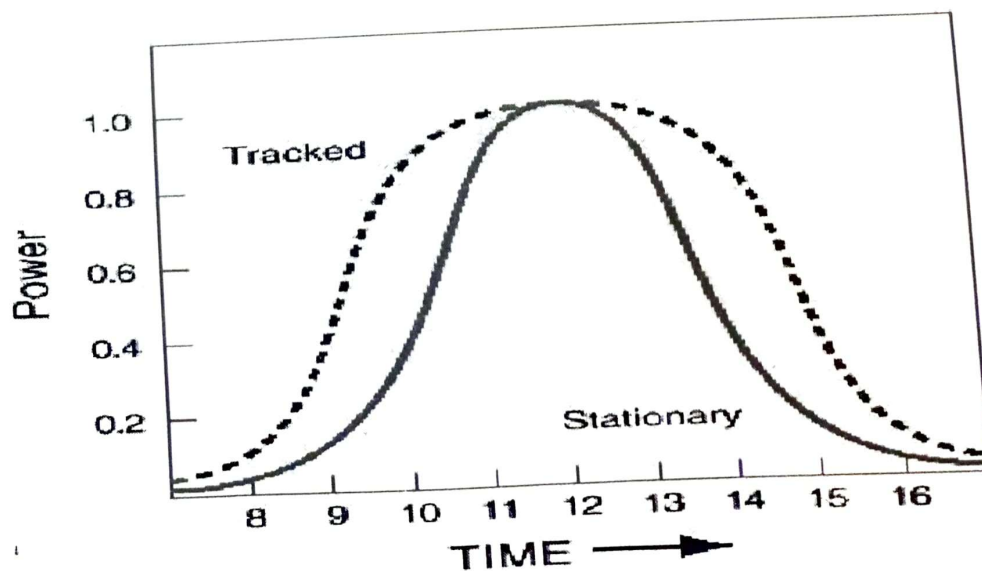
- It's an electrically operated switch used for switching of large currents.
- Uses an electromagnet to operate a switching mechanism mechanically.



PROGRAMMABLE LOGIC CONTROLLER

- A **programmable logic controller (PLC)** or **programmable controller** is an industrial digital controller which has been ruggedized and adapted for the control of manufacturing processes, such as assembly lines, or robotic devices, or any activity that requires high reliability, ease of programming and process fault diagnosis.

COMPARISON OF SOLAR ENERGY



ADVANTAGES

- ❑ Simple
- ❑ Low cost
- ❑ Eco-Friendly
- ❑ We can monitor directly using PC Tracking accuracy is more
- ❑ Reduce the usage power of power grid.

DISADVANTAGE

- ❑ Solar electricity is not available at night and is less available in cloudy weather conditions. Therefore, a storage or complimentary power system is required.
- ❑ Limited power density.
- ❑ Solar cells produce DC which must be converted to AC when used in currently existing distribution grids

CONCLUSION

- ❑ The invention of Solar Tracking System helps us improve performance of PV solar system in a simple way
- ❑ Used relative method of sunlight strength.
- ❑ Established a model of automatic tracking system to keep vertical contact between solar panels and sunlight.
- ❑ Improved the utilization rate of solar energy and efficiency of photovoltaic power generation system.